

Data Limitations & Analysis Reflection

Analyzing self-submitted data from platforms like Grad Café introduces significant selection and voluntary response bias that must be factored into any downstream interpretation. Because participation is completely unmoderated and optional, the resulting dataset represents a non-random sample heavily skewed toward applicants with extreme outcomes—primarily those with top-tier metrics receiving early acceptances or those voicing early rejections. Consequently, the sample lacks statistical representativeness across the broader applicant pool. This bias is clearly visible in our analysis results: for instance, Question 4 yielded an average GPA of 3.82 for Fall 2026 American applicants. In practice, this metric is artificially high and reflects a self-reporting tendency where high-performing candidates are far more inclined to share their scores publicly, leading to an overestimation of actual admission baselines.

Beyond sampling bias, the anonymous and unverified nature of the inputs creates issues with data missingness, consistency, and overall reliability. Optional fields such as GRE scores, sub-scores, and exact decision dates are frequently left blank, limiting the viability of deeper multi-variable regressions. Furthermore, raw user entries lack standardized data validation, resulting in duplicate submissions, typos, and fragmented naming conventions. This challenge directly surfaces when comparing the raw string queries to the normalized LLM fields in Question 9. Querying raw program text for Computer Science PhD acceptances undercounted total admissions compared to the LLM-classified field simply because applicants entered variations like 'CS', 'CompSci', or 'Ph.D. in Computer Science'. Without programmatic normalization or LLM entity extraction, reliance on unvalidated, self-reported text strings introduces structural noise that distorts baseline counts and decision timelines.